

**ISA 345 GROUP PROJECT
SECTION D**



AIRPORT DATABASE

How to Optimize Scheduling and Minimize Inefficiencies

Prepared For

Joseph Nwankpa

Prepared By

Kathleen Kennedy, Paige Charlton,
Megan Watson, Kayla McNab

Project Overview

Airports are highly complex, interconnected operations where the alignment and coordination of aircraft routing and crew assignments, and pairings are essential for maintaining a timely, smooth, and efficient system. Unfortunately, many airports rely on outdated and old database systems that make it difficult to maintain the high level of operations needed. As a result of these outdated systems, airports often experience scheduling mistakes, manual workarounds, and operational delays that affect the entire system, including the customers.

These inefficiencies can have significant consequences. One scheduling error can force last-minute crew reassignments, which can cause an aircraft to miss its departure time. Meanwhile, crew pairing issues, like a pilot or flight attendant being scheduled for a flight they can not work, require solutions fast that usually involve manual data entry, human intervention, and crew repositioning, which can be very costly. These types of coordination failures are well-documented across major airports and have led to public complaints, negative media coverage, and reduced customer satisfaction.

When routing and crew data fail to operate effectively, airports and airlines face several issues. These include, but are not limited to, increased operational costs, customer dissatisfaction, loss of revenue, and strained workforce relationships. Given the widespread of these problems, it's clear that the current systems that many airports use are not fully effective for the high level of coordination and operation needed. With this in mind, we built a database designed to improve communication and operations for airports around the country. This database will include **customers, customer satisfaction, airports, aircraft, routes, maintenance schedule, flights, delay events, crew assignments, crew members, and crew schedule.**

The database supports numerous critical transactions, including assigning crew members to flights, updating aircraft maintenance records, and recording customer satisfaction responses. It also stores flight schedules, logs delay events, and manages crew availability. These transactions ensure that all operational data remains accurate, accessible, and aligned across the entire airport system.

Database Design and Implementation

Entities and Relationships

Customers:

The Customers entity includes data about each customer that has ever come to the airport. The CustomerID is the identifier for each customer. The entity also includes information regarding the customers full name, email address, and phone number.

Ratings:

The Ratings entity includes information regarding the customers ratings to the airport. The Rating ID is the identifier used to define the rating and connect it to a customer ID. The entity provides information on rating, customer ID, and Airport Code.

- a) Ratings can only have one Customer Per Rating
- b) Ratings can only have one Airport per Rating

Airports:

The Airport entity contains data about each airport. Airport code acts as an identifier linking the airport to the airport name and city. This entity contains data on airport name, city, time zone, number of gates, runway count, and max flights per an hour.

- a) An airport could have multiple routes

Routes:

The Routes entity contains information regarding each route an airplane takes. The identifier is Route ID which links the route to the airline and airport. This entity contains data on original airport code, destination airport code, standard flight time, and distance in miles.

- a) A route has one airport
- b) A route has multiple flights

Aircraft:

The Aircraft entity stores detailed information about every airplane in the system. AircraftID functions as the unique identifier that distinguishes each aircraft from the others. In addition to this primary key, the entity includes attributes that describe the aircraft type, its total seating capacity, and its current operational or functional status.

- a) An aircraft could have many flights
- b) An aircraft could have multiple maintenance schedules

Flights:

The Flights entity contains information about every flight arriving at or departing from the airport. FlightID serves as the unique identifier for each flight. The entity also records the associated route ID, the aircraft assigned to the flight, the scheduled departure and arrival times, and the actual departure and arrival times.

- a) A flight has one aircraft
- b) A flight could have multiple delay events
- c) A flight has multiple crew assignments
- d) A flight has one route

CrewAssignments:

The Crew Assignments entity stores information about crew assignments for each flight. AssignmentID serves as the unique identifier for each assignment. The entity also includes the IDs of the crew members and the corresponding flight ID.

- a) A crew assignment has one flight
- b) A crew assignment has multiple crew members

CrewMembers:

The Crew Members entity contains information about each crew member. CrewID serves as the unique identifier for each individual. The entity also includes the crew member's first and last names, as well as their role.

- a) A crew member could have one crew assignment
- b) A crew member could have multiple crew schedules

CrewSchedule:

The Crew Schedule entity stores information about each crew member's work schedule. ScheduleID serves as the unique identifier for each schedule. The entity also includes the crew member's ID, the start and end times of their shift, and their current status.

- a) A crew schedule could have one crew member

DelayEvents:

The Delay Events entity stores information about events that may impact flights. EventID serves as the unique identifier for each event. The entity also includes the associated flight ID, the type of event, and the delay duration in minutes.

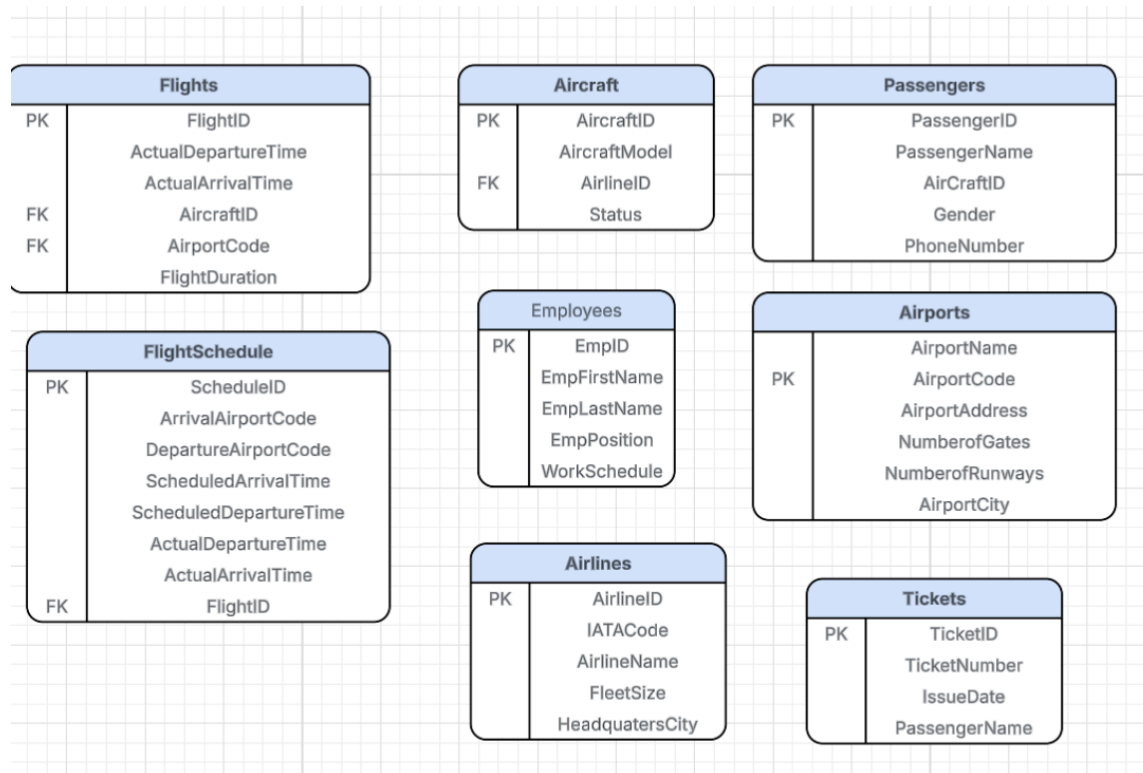
- a) A delay event could have multiple impacted flights

Maintenance:

The Maintenance entity contains information about the maintenance schedule for each aircraft. MaintenanceID serves as the unique identifier for each maintenance event. The entity also includes the associated aircraft ID, the scheduled start and end times, the type of maintenance, and the actual start and end times.

- a) A maintenance event may be associated with a single aircraft

Original Logical Model

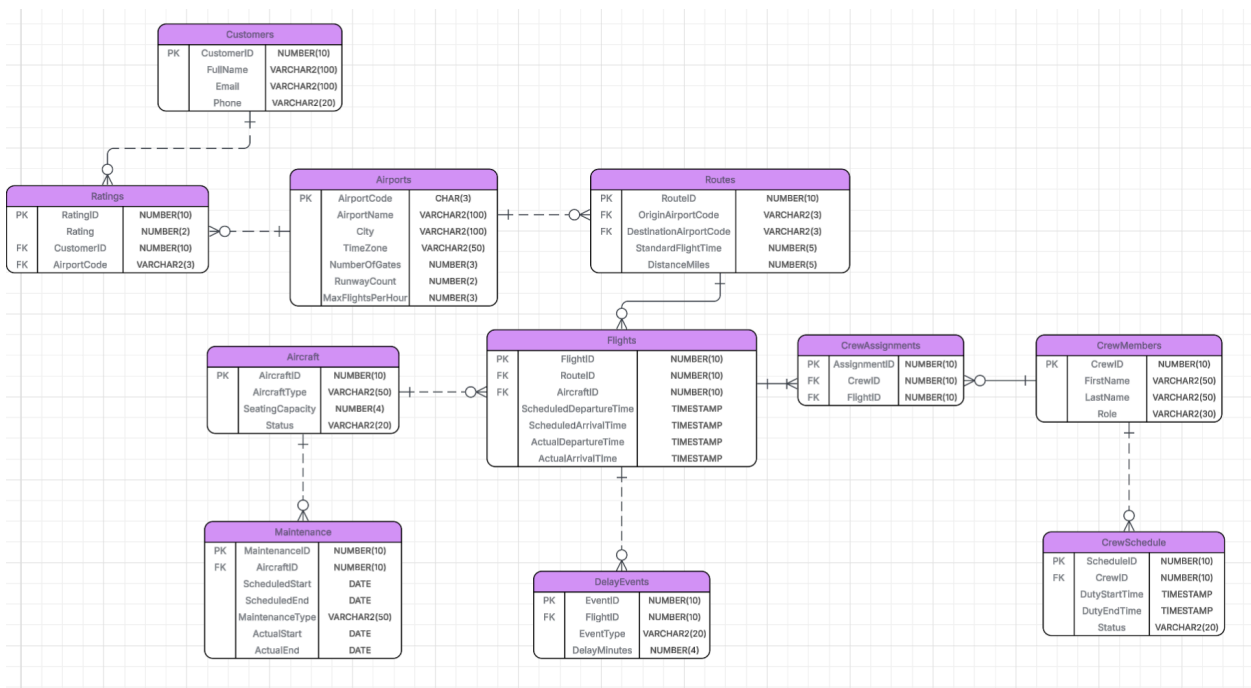


Issues with Logical Model

This original logical model has a few normalization issues that need to be addressed. Beginning with the functional dependencies. Beginning with functional dependencies, passenger information was in more than one table, such as PassengerName being included in both the Passengers and Tickets tables. AircraftID was placed in the Passengers table, which incorrectly tied a passenger directly to an aircraft instead of linking them through a flight. In addition, the original model displayed above did not satisfy the first normal form. For example, in the Flights entity there could have been multiple values for attributes such as ActualDepartureTime and ActualArrivalTime if scheduled and actual events were not separated properly. Also, our original logical model did not satisfy the 2nd normal form because we did not eliminate functional dependencies. As previously mentioned, PassengerName was repeated across a couple entities and it would have caused redundancies within our model. After we realized these errors, we determined the following **PassengerID → PassengerName, Gender, PhoneNumber; TicketID → TicketNumber, IssueDate, PassengerID; AircraftID → AircraftModel, SeatingCapacity, Status; FlightID → ActualDepartureTime, ActualArrivalTime, AircraftID; and ScheduleID → ScheduledDepartureTime, ScheduledArrivalTime, DepartureAirportCode, ArrivalAirportCode**. There were more functional dependencies that affected the model and created redundancies; however, these were the main ones that we thought caused the most errors.

We used these functional dependencies and the issues from the original model to create our second and finalized logical model.

Second and Finalized Logical Model



Improvements with Finalized Model

The finalized logical model significantly strengthens the overall structure of our database by correcting the major issues that appeared in the first version. In the original model, several attributes were repeated across different tables, such as passenger details and flight-related data, which would have caused inconsistencies and made updates difficult. Some relationships were also misplaced, like tying passengers directly to aircraft instead of connecting them through flights, which didn't accurately reflect how airport operations actually work. We also noticed that certain entities combined scheduled and actual values in ways that violated the First Normal Form, making it difficult to track accurate operational timelines.

In the finalized version, we reorganized the tables based on clear functional dependencies and ensured each entity stores only the information directly related to it. We also strengthened the model by incorporating key operational components, including delay events, maintenance schedules, crew assignments, and crew work hours. These additions make the database more realistic and usable for real airport processes. As a result, the finalized model is cleaner, fully normalized, and better aligned with how airports manage routing, scheduling, and operational decision-making. This structure facilitates smoother coordination among flights, crews, and aircraft, thereby reducing the risk of errors that could impact airport performance.

Operational SQL

1. Subquery

```
SELECT MAINTENANCE.AircraftID, AIRCRAFT.Model, (MAINTENANCE.ActualEnd -  
MAINTENANCE.ActualStart) AS MaintenanceDuration_Days  
FROM MAINTENANCE  
JOIN AIRCRAFT ON MAINTENANCE.AircraftID = AIRCRAFT.AircraftID  
WHERE (MAINTENANCE.ActualEnd - MAINTENANCE.ActualStart) = (SELECT  
MAX(ActualEnd - ActualStart) FROM MAINTENANCE);
```

This query allows you to find the aircraft that has been in maintenance the longest. By including both the AircraftID and the Aircraft Model, you can gain clear insight into what specific aircraft required the most attention by the maintenance team. By using the actual start and end date to calculate the maintenance duration time it highlights exactly how long the aircraft was under maintenance.

2. Inner join

```
SELECT CrewAssignments.AssignmentID, CrewMembers.FirstName,  
CrewMembers.LastName AS CrewMemberName, Flights.FlightID,  
Flights.ScheduledDepartureTime, Flights.ScheduledArrivalTime, CrewSchedule.DutyStartTime,  
CrewSchedule.DutyEndTime  
FROM CrewAssignments  
JOIN CrewMembers ON CrewAssignments.CrewID = CrewMembers.CrewID  
JOIN Flights ON CrewAssignments.FlightID = Flights.FlightID  
JOIN CrewSchedule ON CrewAssignments.CrewID = CrewSchedule.CrewID  
WHERE Flights.ScheduledDepartureTime < CrewSchedule.DutyStartTime OR  
Flights.ScheduledArrivalTime > CrewSchedule.DutyEndTime;
```

This query can be used to find crew members assigned to flights scheduled outside their duty hours. By displaying the crew members' names alongside the arrival and departure time of the flights, you can gain a clear view of what assignments may violate labor regulations or policies. The inclusion of each crew member's start and end time of duty lays out exactly when their shift is and can see if there is any overlap. It is particularly valuable for scheduling at the airport as it helps make sure that the schedules are following the crew rest requirements and prevents overscheduling. This leads to safer and more efficient flight operations.

3. Outer join

```
SELECT ROUTES.RouteID, ROUTES.OriginAirportCode, ROUTES.DestinationAirportCode,  
FLIGHTS.FlightID, FLIGHTS.ScheduledDepartureTime  
FROM ROUTES  
LEFT JOIN FLIGHTS ON ROUTES.RouteID = FLIGHTS.RouteID  
ORDER BY ROUTES.RouteID;
```

This query shows every single route and flightid that occurs, even if no flights are currently scheduled. This is because of the left join between routes and flights that ensures that all routes are available to view even if flightid is null. This is especially valuable for operations planning because it highlights unused or underutilized routes that may need more scheduling attention.

4. GROUP BY clause without the HAVING clause:

```
SELECT Aircraft.AircraftID, Aircraft.Model, COUNT(Flights.FlightID) AS  
TotalFlightsOperated  
FROM Aircraft  
JOIN Flights ON Aircraft.AircraftID = Flights.AircraftID  
GROUP BY Aircraft.AircraftID, Aircraft.Model  
ORDER BY TotalFlightsOperated DESC;
```

This query calculates how frequently each aircraft has been used by calculating the total number of flights operated by every aircraft in the system. By grouping the results by AircraftID and model the query provides a clear summary of the patterns across the aircrafts available. This is crucial for maintenance schedule and rotation planning so one flight isn't being overused and another being underutilized.

5. GROUP BY clause together with the HAVING clause:

```
SELECT Model, SUM(Capacity) AS TotalModelCapacity  
FROM AIRCRAFT  
GROUP BY Model  
HAVING SUM(Capacity) > 300  
ORDER BY TotalModelCapacity DESC;
```

This query identifies which aircraft models contribute the most overall seating capacity within the fleet (collection of aircrafts). By grouping aircraft by model and summing the

capacity, the query highlights how much a total passenger capacity each different model can support. The capacity being greater than 300 seats allows airport workers analyzing this to focus on the largest and most impactful aircraft. This information is useful for planning high capacity routes where there is a high demand from customers.

6. Define relevant INDEXES

a) `CREATE INDEX IX_DelayEvents_FlightID ON DELAYEVENTS (FlightID);`

This index was created to allow quicker access to delay records associated with a specific flight. Since delay information is frequently analyzed alongside flight operations, queries often reference the flightid when trying to come up with the cause of delays or the total delay time. The index key is the flight id, which enables the system to locate all delay events tied to a specific flight a lot faster.

b) `CREATE INDEX IX_Routes_Airports ON ROUTES (OriginAirportCode, DestinationAirportCode);`

This index was created to improve performance when retrieving route information based on airport destination and origin pairs. The origin airport code and destination airport code are commonly added together in a query so by indexing these two attributes as a composite key, the database can more efficiently match routes without looking at the entire routes table.

c) `CREATE INDEX IX_CrewSchedule_CrewID ON CREWSCHEDULE (CrewID);`

This index was created to help retrieve crew scheduling information quickly and consistently. The crewid is frequently used when determining a crew member's assigned schedule, checking availability, or preventing scheduling conflicts. Because these tasks require repeatedly accessing the crew schedule for individual crew members, indexing crewid greatly speeds up the lookup time.

7-8. Define a relevant VIEW and Query that created VIEW:

```
CREATE VIEW LIMITED_FLIGHTINFO AS
SELECT FlightID, RouteID, ScheduledDepartureTime, ScheduledArrivalTime, Status
FROM FLIGHTS;
```

```
SELECT LIMITED_FLIGHTINFO.FlightID,
       ROUTES.DepartureAirport,
```

```

    ROUTES.ArrivalAirport,
    LIMITED_FLIGHTINFO.ScheduledDepartureTime,
    LIMITED_FLIGHTINFO.Status
FROM LIMITED_FLIGHTINFO
JOIN ROUTES
    ON LIMITED_FLIGHTINFO.RouteID = ROUTES.RouteID
WHERE LIMITED_FLIGHTINFO.Status = 'On Time';

```

This view was created to limit the amount of detailed flight information that certain members of the organization can access. By displaying only basic flight details such as flightid, route, scheduled times, and status, the view prevents users from seeing more advanced operational details stored in the flights table. In an airline setting, detailed operational metrics, maintenance notes, crew schedules, and delay duration may be restricted to authorized personnel who handle these matters. Using this limited view helps ensure that private information and data remains secure and is only accessible to staff members who need it.

9. Define a relevant PROCEDURE:

```

CREATE OR REPLACE PROCEDURE UpdateMaintenanceEnd (MaintenanceID IN
NUMBER, NewActualEnd IN DATE)
AS
BEGIN
    UPDATE MAINTENANCE SET ActualEnd = NewActualEnd WHERE MaintenanceID =
MaintenanceID;

```

This procedure is used to update the ActualEnd date for existing maintenance events. It automatically calculates the duration for reporting purposes. When a database is executed, it takes two inputs: maintenanceid of the event that needs to be updated and the new actual end date. The procedure then locates the correct maintenance record and updates its ActualEnd field accordingly. This helps make sure that any changes to the maintenance timing are accurately reflected in the system, allowing the organization to maintain correct records for reporting and scheduling purposes.

10. Define a relevant TRIGGER

```

CREATE OR REPLACE TRIGGER CheckFlightTimes
BEFORE INSERT OR UPDATE ON Flights
FOR EACH ROW
BEGIN

```

```
IF :NEW.ActualArrivalTime < :NEW.ActualDepartureTime THEN  
RAISE_APPLICATION_ERROR(-20010,'Arrival time cannot be earlier than departure time.');
```

END IF;
END;

This trigger runs before each insert or update, checking the new values being entered. If the system detects that the arrival time is earlier than the departure time, it raises a custom Oracle error and stops the transaction from completing. This ensures that all flight records follow a logical timeline and prevents invalid or contradictory scheduling data from being saved in the database.

Conclusions & Lessons Learned

The database we created allows users to efficiently access, store, and retrieve information about airports and their operations. By organizing data accurately and eliminating redundancies, the system supports better decision-making and highlights areas in need of attention. Its built-in trigger safeguards data integrity by preventing arrival times from being entered earlier than departure times, reducing human error before it reaches the system. The view functionality also ensures appropriate access control, helping maintain security and preventing accidental modifications. Throughout development, we applied normalization to properly structure entities and resolve redundancies, resulting in a well-organized and reliable database.

Throughout the project, several design decisions shaped the final system. We originally planned a simpler structure, but we expanded the database to include customer satisfaction, crew assignments, and detailed crew scheduling after recognizing how essential these elements are for understanding airport performance and staffing efficiency. Incorporating these tables required revisiting our original entity relationships and refining our normalization process to ensure accurate connections across the system. One challenge we encountered was maintaining strict data accuracy, which led us to implement the arrival-before-departure trigger to prevent inconsistent flight information. Despite these changes, we successfully met our project objectives by delivering a well-organized, accurate, and comprehensive database that supports better operations and decision-making.

Because airports are major hubs of travel and transportation, they depend on precise and timely information. Operational failures can be costly and significantly affect customer satisfaction. Our database supports smoother operations by providing accurate, comprehensive data across all core tables—including customers, customer satisfaction, airports, aircraft, routes, maintenance schedules, flights, delay events, crew assignments, crew members, and crew schedules. This structure also greatly improves crew scheduling, allowing managers to see crew availability, track assigned shifts, and ensure that flights are staffed appropriately and in compliance with

regulations. Additionally, managers can analyze key metrics such as overall fleet seating capacity and aircraft usage frequency, along with many other operational insights. For these reasons, our database is an essential tool for maintaining up-to-date operations and continuously improving airport efficiency and effectiveness.

Appendices

Create Tables SQL

```
CREATE TABLE AIRPORTS (  
    AirportCode VARCHAR2(3) CONSTRAINT AirportsPK PRIMARY KEY,  
    AirportName VARCHAR2(100) NOT NULL,  
    City VARCHAR2(100) NOT NULL,  
    Country VARCHAR2(100) NOT NULL  
);
```

```
CREATE TABLE CUSTOMERS (  
    CustomerID NUMBER(10) CONSTRAINT CustomersPK PRIMARY KEY,  
    FirstName VARCHAR2(50) NOT NULL,  
    LastName VARCHAR2(50) NOT NULL,  
    Email VARCHAR2(100) NOT NULL  
);
```

```
CREATE TABLE CREWMEMBERS (  
    CrewID NUMBER(10) CONSTRAINT CrewMembersPK PRIMARY KEY,  
    FirstName VARCHAR2(50) NOT NULL,  
    LastName VARCHAR2(50) NOT NULL,  
    Role VARCHAR2(30) NOT NULL  
);
```

```
CREATE TABLE AIRCRAFT (  
    AircraftID NUMBER(10) CONSTRAINT AircraftPK PRIMARY KEY,  
    Model VARCHAR2(50) NOT NULL,  
    Capacity NUMBER(4) NOT NULL  
);
```

```
CREATE TABLE ROUTES (  
    RouteID NUMBER(10) CONSTRAINT RoutesPK PRIMARY KEY,  
    OriginAirportCode VARCHAR2(3)  
        CONSTRAINT RoutesOriginFK REFERENCES AIRPORTS(AirportCode) NOT NULL,  
    DestinationAirportCode VARCHAR2(3)  
        CONSTRAINT RoutesDestinationFK REFERENCES AIRPORTS(AirportCode) NOT  
NULL,  
    StandardFlightTime NUMBER(5) NOT NULL,  
    DistanceMiles NUMBER(5) NOT NULL
```

);

```
CREATE TABLE FLIGHTS (  
    FlightID NUMBER(10) CONSTRAINT FlightsPK PRIMARY KEY,  
    RouteID NUMBER(10)  
        CONSTRAINT FlightsRouteFK REFERENCES ROUTES(RouteID) NOT NULL,  
    AircraftID NUMBER(10)  
        CONSTRAINT FlightsAircraftFK REFERENCES AIRCRAFT(AircraftID) NOT NULL,  
    ScheduledDepartureTime TIMESTAMP NOT NULL,  
    ScheduledArrivalTime TIMESTAMP NOT NULL,  
    ActualDepartureTime TIMESTAMP NOT NULL,  
    ActualArrivalTime TIMESTAMP NOT NULL  
);
```

```
CREATE TABLE RATINGS (  
    RatingID NUMBER(10) CONSTRAINT RatingsPK PRIMARY KEY,  
    Rating NUMBER(2) NOT NULL,  
    CustomerID NUMBER(10)  
        CONSTRAINT RatingsCustomerFK REFERENCES CUSTOMERS(CustomerID) NOT  
NULL,  
    AirportCode VARCHAR2(3)  
        CONSTRAINT RatingsAirportFK REFERENCES AIRPORTS(AirportCode) NOT NULL  
);
```

```
CREATE TABLE DELAYEVENTS (  
    EventID NUMBER(10) CONSTRAINT DelayEventsPK PRIMARY KEY,  
    FlightID NUMBER(10)  
        CONSTRAINT DelayEventsFlightFK REFERENCES FLIGHTS(FlightID) NOT NULL,  
    EventType VARCHAR2(20) NOT NULL,  
    DelayMinutes NUMBER(4) NOT NULL  
);
```

```
CREATE TABLE CREWASSIGNMENTS (  
    AssignmentID NUMBER(10) CONSTRAINT CrewAssignmentsPK PRIMARY KEY,  
    CrewID NUMBER(10)  
        CONSTRAINT CrewAssignmentsCrewFK REFERENCES CREWMEMBERS(CrewID)  
NOT NULL,  
    FlightID NUMBER(10)  
        CONSTRAINT CrewAssignmentsFlightFK REFERENCES FLIGHTS(FlightID) NOT  
NULL
```

);

```
CREATE TABLE CREWSCHEDULE (  
    ScheduleID NUMBER(10) CONSTRAINT CrewSchedulePK PRIMARY KEY,  
    CrewID NUMBER(10)  
        CONSTRAINT CrewScheduleCrewFK REFERENCES CREWMEMBERS(CrewID) NOT  
    NULL,  
    DutyStartTime TIMESTAMP NOT NULL,  
    DutyEndTime TIMESTAMP NOT NULL,  
    Status VARCHAR2(20) NOT NULL  
);
```

Insert Statements SQL

CUSTOMERS

```
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (1,  
'Emily', 'Johnson', 'emily.johnson@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (2,  
'Michael', 'Smith', 'm.smith@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (3, 'Sarah',  
'Davis', 'sarah.davis@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (4,  
'Daniel', 'Harris', 'dan.harris@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (5,  
'Chloe', 'Martinez', 'chloe.m@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (6,  
'Joshua', 'Brown', 'jbrown@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (7, 'Ava',  
'Thompson', 'ava.t@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (8, 'Ethan',  
'Wilson', 'ewilson@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (9,  
'Olivia', 'Clark', 'olivia.c@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (10,  
'William', 'Lewis', 'wlewis@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (11,  
'Sophia', 'Walker', 'swalker@example.com');  
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (12,  
'James', 'Hall', 'jhall@example.com');
```

```
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (13,
'Isabella', 'Young', 'iyoung@example.com');
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (14,
'Benjamin', 'King', 'bking@example.com');
INSERT INTO CUSTOMERS (CustomerID, FirstName, LastName, Email) VALUES (15, 'Mia',
'Scott', 'mia.scott@example.com');
```

AIRPORTS

```
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('CVG',
'Cincinnati/Northern Kentucky Intl', 'Cincinnati', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('LAX', 'Los
Angeles Intl', 'Los Angeles', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('ORD',
'O'Hare Intl', 'Chicago', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('ATL',
'Atlanta Hartsfield-Jackson', 'Atlanta', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('DFW',
'Dallas/Fort Worth Intl', 'Dallas', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('JFK', 'John
F. Kennedy Intl', 'New York', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('BOS',
'Logan Intl', 'Boston', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('MIA',
'Miami Intl', 'Miami', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('SEA',
'Seattle-Tacoma Intl', 'Seattle', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('DEN',
'Denver Intl', 'Denver', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('PHX',
'Phoenix Sky Harbor', 'Phoenix', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('SFO', 'San
Francisco Intl', 'San Francisco', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('DTW',
'Detroit Metro', 'Detroit', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('MSP',
'Minneapolis-Saint Paul', 'Minneapolis', 'USA');
INSERT INTO AIRPORTS (AirportCode, AirportName, City, Country) VALUES ('CLT',
'Charlotte Douglas Intl', 'Charlotte', 'USA');
```


RATINGS

```
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (1, 5, 1, 'CVG');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (2, 4, 2, 'LAX');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (3, 3, 3, 'ORD');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (4, 5, 4, 'ATL');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (5, 2, 5, 'DFW');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (6, 4, 6, 'JFK');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (7, 5, 7, 'BOS');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (8, 3, 8, 'MIA');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (9, 4, 9, 'SEA');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (10, 1, 10, 'DEN');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (11, 5, 11, 'PHX');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (12, 4, 12, 'SFO');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (13, 3, 13, 'DTW');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (14, 5, 14, 'MSP');
INSERT INTO RATINGS (RatingID, Rating, CustomerID, AirportCode) VALUES (15, 4, 15, 'CLT');
```

AIRCRAFT:

```
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (1, 'Boeing 737-800', 160);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (2, 'Airbus A320', 150);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (3, 'Boeing 777-300', 350);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (4, 'Airbus A321', 185);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (5, 'Boeing 787-9', 290);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (6, 'Airbus A319', 132);
```

```
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (7, 'Boeing 757-200', 180);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (8, 'Airbus A330', 250);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (9, 'Boeing 767-300', 260);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (10, 'Boeing 737 MAX',
178);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (11, 'Airbus A350', 325);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (12, 'Boeing 747-400',
416);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (13, 'Airbus A220', 120);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (14, 'Boeing 717', 110);
INSERT INTO AIRCRAFT (AircraftID, Model, Capacity) VALUES (15, 'Embraer E175', 76);
```

ROUTES

```
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (1, 'CVG', 'LAX', 240, 1900);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (2, 'LAX', 'CVG', 240, 1900);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (3, 'ATL', 'JFK', 130, 760);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (4, 'JFK', 'ATL', 130, 760);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (5, 'ORD', 'DFW', 150, 800);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (6, 'DFW', 'ORD', 150, 800);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (7, 'CVG', 'MIA', 160, 950);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (8, 'MIA', 'CVG', 160, 950);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (9, 'SFO', 'SEA', 120, 680);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (10, 'SEA', 'SFO', 120, 680);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (11, 'DEN', 'PHX', 110, 600);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (12, 'PHX', 'DEN', 110, 600);
```

```

INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (13, 'CLT', 'BOS', 130, 720);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (14, 'BOS', 'CLT', 130, 720);
INSERT INTO ROUTES (RouteID, OriginAirportCode, DestinationAirportCode,
StandardFlightTime, DistanceMiles) VALUES (15, 'DTW', 'MSP', 120, 700);

```

FLIGHTS

```

INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (1, 1, 1,
TO_TIMESTAMP('2025-03-01 08:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-01 12:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-01 08:05','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-01 12:10','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (2, 2, 2,
TO_TIMESTAMP('2025-03-01 14:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-01 18:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-01 14:01','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-01 18:05','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (3, 3, 3,
TO_TIMESTAMP('2025-03-02 09:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-02 11:10','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-02 09:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-02 11:12','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (4, 4, 4,
TO_TIMESTAMP('2025-03-02 12:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-02 14:10','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-02 12:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-02 14:09','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (5, 5, 5,
TO_TIMESTAMP('2025-03-03 06:30','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-03 09:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-03 06:45','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-03 09:20','YYYY-MM-DD HH24:MI'));

```

```

INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (6, 6, 6,
TO_TIMESTAMP('2025-03-03 10:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-03 12:30','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-03 10:02','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-03 12:28','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (7, 7, 7,
TO_TIMESTAMP('2025-03-04 08:30','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-04 11:10','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-04 08:40','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-04 11:22','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (8, 8, 8,
TO_TIMESTAMP('2025-03-04 15:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-04 17:40','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-04 15:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-04 17:35','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (9, 9, 9,
TO_TIMESTAMP('2025-03-05 09:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-05 11:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-05 09:15','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-05 11:20','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (10, 10, 10,
TO_TIMESTAMP('2025-03-05 13:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-05 15:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-05 13:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-05 15:02','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (11, 11, 11,
TO_TIMESTAMP('2025-03-06 07:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-06 09:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-06 07:10','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-06 09:19','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (12, 12, 12,
TO_TIMESTAMP('2025-03-06 18:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-06 20:00','YYYY-MM-DD HH24:MI'),

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```

TO_TIMESTAMP('2025-03-06 18:02','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-06 20:10','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (13, 13, 13,
TO_TIMESTAMP('2025-03-07 11:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-07 13:10','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-07 11:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-07 13:12','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (14, 14, 14,
TO_TIMESTAMP('2025-03-07 17:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-07 19:10','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-07 17:10','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-07 19:21','YYYY-MM-DD HH24:MI'));
INSERT INTO Flights (FlightID, RouteID, AircraftID, ScheduledDepartureTime,
ScheduledArrivalTime, ActualDepartureTime, ActualArrivalTime) VALUES (15, 15, 15,
TO_TIMESTAMP('2025-03-08 08:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-08 10:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-08 08:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-08 10:03','YYYY-MM-DD HH24:MI'));

```

DELAYEVENTS

```

INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (1, 1,
'Weather', 10);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (2, 2,
'Crew', 5);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (3, 5,
'Mechanical', 20);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (4, 7,
'Weather', 12);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (5, 9,
'Technical', 15);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (6, 11,
'Crew', 10);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (7, 12,
'Weather', 8);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (8, 14,
'Runway', 11);

```

```

INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (9, 3,
'Security', 6);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (10, 4,
'Late Arrival', 9);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (11, 6,
'Fueling', 7);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (12, 8,
'Catering', 5);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (13, 10,
'Weather', 4);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (14, 13,
'Gate Change', 6);
INSERT INTO DelayEvents (EventID, FlightID, EventType, DelayMinutes) VALUES (15, 15,
'Baggage', 3);

```

CREWMEMBERS

```

INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (1, 'John',
'Carter', 'Pilot');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (2, 'Emily',
'Rogers', 'Co-Pilot');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (3, 'Mark',
'Thompson', 'Flight Attendant');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (4, 'Sarah',
'Mitchell', 'Flight Attendant');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (5, 'David',
'Lee', 'Pilot');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (6, 'Laura',
'Adams', 'Co-Pilot');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (7,
'Nathan', 'Brooks', 'Flight Attendant');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (8, 'Rachel',
'King', 'Flight Attendant');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (9, 'Chris',
'Wright', 'Pilot');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (10, 'Anna',
'Torres', 'Co-Pilot');

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```

INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (11, 'Jacob',
'Perry', 'Flight Attendant');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (12,
'Megan', 'Hughes', 'Flight Attendant');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (13, 'Ryan',
'Murphy', 'Pilot');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (14,
'Grace', 'Foster', 'Co-Pilot');
INSERT INTO CREWMEMBERS (CrewID, FirstName, LastName, Role) VALUES (15,
'Kevin', 'Ward', 'Flight Attendant');

```

CREWSCHEDULE

```

INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (1, 1, TO_TIMESTAMP('2025-03-01 06:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-01 14:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (2, 2, TO_TIMESTAMP('2025-03-01 06:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-01 14:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (3, 3, TO_TIMESTAMP('2025-03-01 07:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-01 15:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (4, 4, TO_TIMESTAMP('2025-03-02 08:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-02 16:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (5, 5, TO_TIMESTAMP('2025-03-02 06:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-02 14:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (6, 6, TO_TIMESTAMP('2025-03-02 06:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-02 14:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (7, 7, TO_TIMESTAMP('2025-03-03 07:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-03 15:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (8, 8, TO_TIMESTAMP('2025-03-03 07:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-03 15:00','YYYY-MM-DD HH24:MI'), 'Completed');

```

```

INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (9, 9, TO_TIMESTAMP('2025-03-04 05:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-04 13:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (10, 10, TO_TIMESTAMP('2025-03-04 05:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-04 13:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (11, 11, TO_TIMESTAMP('2025-03-05 09:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-05 17:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (12, 12, TO_TIMESTAMP('2025-03-05 09:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-05 17:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (13, 13, TO_TIMESTAMP('2025-03-06 06:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-06 14:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (14, 14, TO_TIMESTAMP('2025-03-06 06:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-06 14:00','YYYY-MM-DD HH24:MI'), 'Completed');
INSERT INTO CREWSCHEDULE (ScheduleID, CrewID, DutyStartTime, DutyEndTime,
Status) VALUES (15, 15, TO_TIMESTAMP('2025-03-07 10:00','YYYY-MM-DD HH24:MI'),
TO_TIMESTAMP('2025-03-07 18:00','YYYY-MM-DD HH24:MI'), 'Completed');

```

CREWASSIGNMENTS

```

INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (1, 1, 1);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (2, 1, 2);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (3, 2, 5);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (4, 2, 6);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (5, 3, 9);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (6, 3, 10);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (7, 4, 13);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (8, 4, 14);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (9, 5, 1);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (10, 5, 2);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (11, 6, 5);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (12, 6, 6);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (13, 7, 9);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (14, 7,
10);

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INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (15, 8, 13);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (16, 8, 14);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (17, 9, 1);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (18, 9, 2);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (19, 10, 5);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (20, 10, 6);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (21, 11, 9);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (22, 11, 10);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (23, 12, 13);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (24, 12, 14);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (25, 13, 1);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (26, 13, 2);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (27, 14, 5);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (28, 14, 6);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (29, 15, 9);
INSERT INTO CREWASSIGNMENTS (AssignmentID, FlightID, CrewID) VALUES (30, 15, 10);

MAINTENANCE

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (1, 5, DATE '2025-01-05', DATE '2025-01-07', 'Engine Check', DATE '2025-01-05', DATE '2025-01-07');
INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (2, 12, DATE '2025-02-01', DATE '2025-02-10', 'Deep Inspection', DATE '2025-02-02', DATE '2025-02-11');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (3, 3, DATE '2025-03-11', DATE '2025-03-14', 'Hydraulic Check', DATE '2025-03-11', DATE '2025-03-14');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (4, 1, DATE '2025-02-20', DATE '2025-02-20', 'Oil Change', DATE '2025-02-20', DATE '2025-02-20');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (5, 2, DATE '2025-01-10', DATE '2025-01-12', 'Landing Gear', DATE '2025-01-10', DATE '2025-01-13');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (6, 4, DATE '2025-03-01', DATE '2025-03-03', 'Cabin Work', DATE '2025-03-01', DATE '2025-03-03');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (7, 6, DATE '2025-02-14', DATE '2025-02-15', 'Electrical Check', DATE '2025-02-14', DATE '2025-02-15');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (8, 7, DATE '2025-04-01', DATE '2025-04-03', 'Fuel System', DATE '2025-04-01', DATE '2025-04-04');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (9, 9, DATE '2025-04-20', DATE '2025-04-25', 'Major Check', DATE '2025-04-20', DATE '2025-04-26');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (10, 11, DATE '2025-05-01', DATE '2025-05-05', 'Engine Overhaul', DATE '2025-05-02', DATE '2025-05-07');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (11, 13, DATE '2025-05-08', DATE '2025-05-09', 'Software Update', DATE '2025-05-08', DATE '2025-05-09');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (12, 14, DATE '2025-03-18', DATE '2025-03-18', 'Tire Replacement', DATE '2025-03-18', DATE '2025-03-18');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (13, 15, DATE '2025-02-23', DATE '2025-02-24', 'System Diagnostics', DATE '2025-02-23', DATE '2025-02-24');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (14, 10, DATE '2025-04-08', DATE '2025-04-09', 'Wing Flap', DATE '2025-04-08', DATE '2025-04-09');

INSERT INTO MAINTENANCE (MaintenanceID, AircraftID, ScheduledStart, ScheduledEnd, MaintenanceType, ActualStart, ActualEnd) VALUES (15, 8, DATE '2025-05-10', DATE '2025-05-12', 'General Check', DATE '2025-05-10', DATE '2025-05-12');